Chapter 19
Extract Sentiment from Customer Reviews: A Better Approach of TF-IDF and BOW-Based Text Classification Using N-Gram Technique

Tonmoy Hasan and Abdul Matin

1 Introduction

Customer satisfaction reflects the success of any business organization and from customer reviews, the organizations can know how much their customers are satisfied [1]. For restaurants or food businesses, it is difficult to make space in the business market without considering customer's feedback. Organization should pay attention to customer's reviews and improve their services in order to allure more and more customers. But it is a very onerous and lengthy job for the human to scrutinize a huge amount of reviews. However, sentiment analysis can easily handle this task. Opinion mining is another introduction of sentiment analysis that aims to attempt the determination of polarity (i.e., positive, negative, or neutral) of text, document, or paragraph [2].

In this paper, we have used three feature analysis techniques such as BOW, TF-IDF and N-Gram. Different N-Gram techniques, for instance, unigram, bigram, trigram, and incorporation of unigram and bigram (Unigram + Bigram), incorporation of bigram and trigram (Bigram + Trigram), and incorporation of unigram, bigram, and trigram (Unigram + Bigram + Trigram) are applied for both BOW and TF-IDF. To get classification result, RF, SVM, NB, and voting ensemble have been employed for sentiment extraction of food reviews. Precision, recall, f1-score, and accuracy are considered as the performance parameter of the techniques.

T. Hasan (✉) · A. Matin
Department of Electrical & Computer Engineering (ECE), Rajshahi University of Engineering & Technology (RUET), Rajshahi 6204, Bangladesh

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1.1 Research Objectives

In this paper, we have proposed an efficient method to classify sentiment of customer reviews to mitigate the following objectives:

(i) Identify the best N-Gram feature for customer sentiment classification.
(ii) Finding out the better one between TF-IDF and BOW feature extraction technique.
(iii) Impacts of voting ensemble classifier on overall performance.

2 Literature Review

Fouad et al. [3] have analyzed sentiment classification and considered for four datasets. BOW, Lexicon-based features (Lex), and Emotion-based features (Emo) have been applied for these datasets. Different machine learning algorithms such as SVM, Logistic Regression (LG), and NB are considered for different combinations of feature extraction techniques. Positive and negative tweets can be detected through their machine learning model. Furthermore, they have also claimed that their proposed system would be effective for marketing, political polarity detection, and reviewing products.

Qamar et al. [4] have considered tweets of various Saudi telecommunication companies such as STC, Zain, etc. for analyzing sentiment. These tweets have been labeled as positive, negative, and neutral through different supervised learning classification algorithms and feature selection formulas. They have gone through NB, K-Nearest Neighbor (KNN), Artificial Neural Network (ANN) as machine learning algorithms and CfsSubsetEvaluation, N-Gram, Info Gain for feature selection techniques.

Elghazaly et al. [5] have used SVM and NB as machine learning techniques in order to determine positive and negative sentiment for Arabic text classification. Political tweets of presidential elections in Egypt 2012 have been analyzed to compare the results of SVM and NB classifiers using several performance parameters (F-measure, Recall, and Precision). They have gone through TF-IDF model as feature extraction technique and their main concerns are accuracy and time. In their research, however, they have found that NB is a more suitable classifier for their main aspects.

Bespalov et al. [6] have analyzed sentiment depending on deep neural network technique for two datasets of online product reviews concerning higher-order N-gram. BOW, N-Gram, and Inverse Document Frequency (IDF) are used for Amazon and TripAdvisor customer reviews. 20,000 and 3000 reviews from Amazon and TripAdvisor ,respectively, have been used and a proportion of 70%/30% has been maintained for train and test purposes.

Zul et al. [7] have explored social media for sentiment analysis conducting K-Means and NB algorithms. Facebook and Twitter are the primary sources for their data. The opinions collected from social media have been labeled by Sentiwordnet. In their research, BOW is used as the feature selection technique. They have found that better accuracy can be achieved through NB alone rather than a combination of NB and K-Means.

3 Proposed Framework and Methodology

The overall view of our proposed system is portrayed in Fig. 1. Initially, data has been collected and preprocessing is accomplished undergoing various steps such as removal of URLs, converting to lowercase, tokenization, removal of expressions, removal of punctuation, removal of stopwords, and stemming. Then the text data has been converted to numeric data through different feature extraction techniques (i.e., TF-IDF, BOW, and N-Gram). After the conversion, individual classifiers such as RF, SVM, NB, and voting ensemble classifiers are applied in an attempt to get classification results. According to the outcomes, business policy and product quality would be updated by business organizations having identified negative reviews.

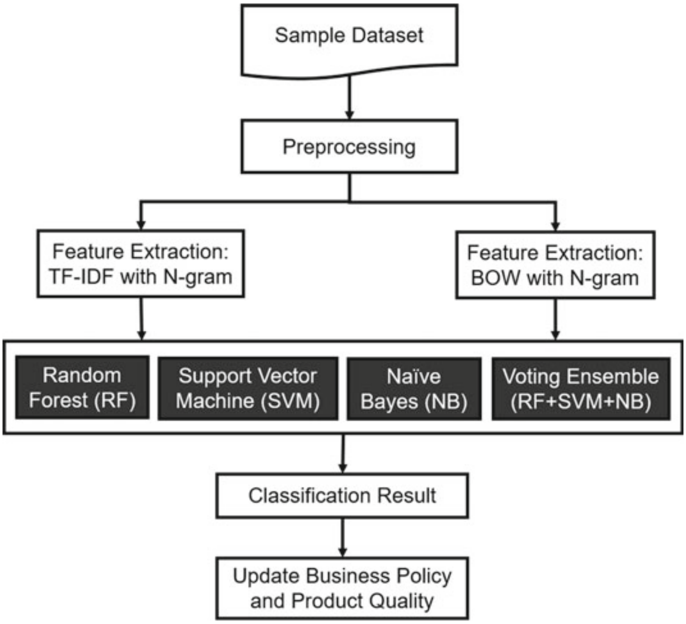

Fig. 1 Proposed system framework

3.1 Dataset

Dataset has been collected from Kaggle named "Sentiment Analysis Classification" [8]. The dataset contains different customer's reviews of foods incorporating total 18,532 reviews for training and testing. We have randomly taken 10,000 reviews for the experiment where 7105 labeled as positive and rest 2895 labeled as negative.

3.2 Preprocessing

Data preprocessing is important to get cleaned data from raw information. Some important steps of preprocessing are:

- Removal of URLs: It is the process of removing all the URLs from the text as they are unnecessary for sentiment classification.
- Converting to lowercase: Converting to lowercase all the text data significantly helps with the consistency of desired outcomes. In natural language processing(NLP), it is one of the most common text processing phases used in research purview.
- Tokenization: It is the step of splitting the reviews into smaller units. These units can be as part like a word is a unit in a sentence and a sentence is a unit in a paragraph. For sentiment analysis, normally word tokenization technique is followed, for example like I enjoyed the food very much to be [I, enjoyed, the, food, very, much].
- Removal of expressions: Expressions don't contain valuable information for sentiment classification. So, removal of expressions is performed as a preprocessing step in order to prepare more clear data.
- Removal of punctuation: Another common preprocessing step in natural language processing is removal of punctuation. It is the process of removing all the punctuation from the text.
- Removal of stopwords: Stopwords are commonly used words in a language like "a", "is", "the", etc. Removal of stopwords from the text doesn't affect the meaning as they don't provide significant information.
- Stemming: Stemming is the preprocessing of shortening the modified words to their word stem, base, or root form. For example, playing can be play, cooked can be cook, disappointed can be disappoint, etc. (Table 1)

3.3 World Cloud Generation

It is a leading process used in NLP for representing text data in a cloud of words [9]. The word clouds have been shown in Figs. 2 and 3 for positive and negative reviews

Table 1 Review before and after Preprocessing

Before preprocessing	After preprocessing
I was very disappointed, I didn't think the coffee lived up to its description. I won't buy it again	Disappoint think coffee live descript buy

Fig. 2 Word cloud for positive reviews

Fig. 3 Word cloud for negative reviews

respectively. In this cloud, the size of the word demonstrates the frequency of the word. More frequently appeared words are larger than less frequent words. It should be noted that only positive words have not appeared in the positive cloud and only negative words have not appeared in the negative cloud rather than the words which are more frequently used in the positive and negative reviews appeared in the positive and the negative word cloud, respectively.

3.4 Feature Extraction

Term Frequency-Inverse Document Frequency (TF-IDF): TF-IDF is one of the most observable feature extraction techniques of NLP which is especially for text modeling [10]. Term frequency indicates the frequency of a specific term or word occurs in a document whereas inverse document frequency indicates the occurrences of the word in all documents. The formula for TF-IDF follows as:

$$tfidf_t = f_{t,d} \times \log \frac{N}{df_t} \tag{1}$$

where,
$tfidf_t$ = weight of term t
$f_{t,d}$ = frequency of term t in document d
N = total number of documents
df_t = number of documents containing the term t.

Bag of Words (BOW): BOW is a very common technique of extracting features from text data [11]. It helps to alter from text to number by counting the occurrence of words within a document and machine learning algorithms work with these numerical data. For example,

- Text 1: They have the best price and best service.
- Text 2: Best service ever.

The vocabulary consists of these eight words: “They”, “have”, “the”, “best”, “price”, “and”, “service”, “ever” (Table 2).

Vector of Text 1: [1, 1, 1, 2, 1, 1, 1, 0]
Vector of Text 2: [0, 0, 0, 1, 0, 0, 1, 1]

N-Gram: In natural language processing, N-Gram technique is a frequently used method where n indicates a continuous number of terms or words[10]. For n=1, it indicates unigram. Similarly, bigram and trigram can be indicated for $n = 2$ and $n = 3$ respectively. If we look upon a text such as “The foods are mind blowing” then N-Gram
Unigram: “The”, “foods”, “are”, “mind”, “blowing”.
Bigram: “The foods”, “foods are”, “are mind”, “mind blowing”.
Trigram: “The foods are”, “foods are mind”, “are mind blowing”.

Table 2 Bag of words representation

	1. They	2. have	3. the	4. best	5. price	6. and	7. service	8. ever
Text 1	1	1	1	2	1	1	1	0
Text 2	0	0	0	1	0	0	1	1

Different N-Gram schemes have been used along with TF-IDF and BOW feature extraction techniques separately in order to fit supervised machine learning models.

3.5 Classifier

Random Forest (RF): RF is a supervised classifier in which decision tree is the elementary unit aiming to perform both classification and regression jobs [12]. It is based on ensemble learning and is made up of a wide number of single decision trees. For the final output, the prediction of each decision tree is considered and depending on the maximum number of votes, result with the highest performance is predicted.

Support Vector Machine (SVM): SVM is one of the most prominent classification algorithms among all the supervised machine learning algorithms that explore data and helps to acknowledge patterns. In this method, binary classification has been accomplished by drawing a hyper-plane that cuts apart the document vectors and maximum distance is maintained between hyper-planes [13]. This is the primary objective of SVM.

Naive Bayes (NB): NB is one of the most powerful and straightforward supervised machine learning algorithms in which classification is accomplished by adopting the Bayes Theorem [14]. It is not only very fast but it also boosts up the performance surprisingly. Moreover, it is a probabilistic algorithm which gives very efficient result for both binary as well as multiclass classification.

Voting Ensemble Classifier: Ensemble classifier is a method of combining multiple individual models in order to secure better performance. Voting ensemble is one of the simplest methods of ensemble learning. In this technique, individual classifier makes a prediction which is considered as a vote and according to the number of majority vote, the final prediction is accomplished [14].

3.6 Performance Parameters

For a classifier, four possible assessments are true positive (TP), false positive (FP), true negative (TN), and false negative (FN). TP denotes the number of reviews that are classified as positive and their actual class is also positive. FP expresses the number of reviews which are classified as positive but their actual class is negative. For the negative reviews, TN and FN are stated similarly. Again, performance parameters, e.g., precision, recall, f1-score, and accuracy are calculated in virtue of these outcomes.

Precision: It is known as the proportion of the number of items exactly identified as positive divided by the total number of items that are classified as positive [12].

Recall: It is the proportion of the number of events exactly identified as positive divided by the total number of exact positive events [12].

F1-Score: It can be achieved from the following equation [12]:

$$\text{F1-score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

Accuracy: Accuracy can be obtained from the following equation [12]:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{3}$$

4 Result and Discussions

In this section of our study, a comparative result has been pictured in order to look out the performances of different techniques. Again, a comprehensive graph analysis has been shown to visualize more deeply. Tables 3 and 4 summarizes accuracy and f1-score respectively for different N-Gram approaches using both TF-IDF and BOW separately. However, the dataset we have used has two labels: positive and negative. In our experiment, total 2500 words are considered as features for all the feature

Table 3 Accuracy of different classifier using features combination

N-Gram	TF-IDF				BOW			
	RF	SVM	NB	Ensemble	RF	SVM	NB	Ensemble
Unigram	0.846	0.861	0.8095	0.8525	0.854	0.8545	0.8505	0.8675
Bigram	0.7835	0.796	0.8015	0.806	0.723	0.7735	0.7945	0.7945
Trigram	0.737	0.746	0.7395	0.7475	0.7135	0.729	0.7275	0.731
Unigram + Bigram	0.8505	0.867	0.8345	0.8575	0.8555	0.851	0.861	0.87
Bigram + Trigram	0.778	0.7925	0.7965	0.801	0.7265	0.7695	0.791	0.793
Unigram + Bigram + Trigram	0.8425	0.864	0.821	0.8555	0.8515	0.8455	0.8615	0.8655

Table 4 Average F1-score of different classifier using features combination

N-Gram	TF-IDF				BOW			
	RF	SVM	NB	Ensemble	RF	SVM	NB	Ensemble
Unigram	0.8575	0.8705	0.84	0.8637	0.86	0.864	0.8579	0.873
Bigram	0.801	0.8345	0.8283	0.8266	0.715	0.8217	0.8137	0.806
Trigram	0.78	0.809	0.8172	0.8103	0.7522	0.8229	0.814	0.8092
Unigram + Bigram	0.8626	0.88	0.861	0.8684	0.8732	0.8608	0.8615	0.8848
Bigram + Trigram	0.7912	0.823	0.8268	0.834	0.722	0.8243	0.8055	0.8114
Unigram + Bigram + Trigram	0.8464	0.871	0.839	0.875	0.858	0.862	0.864	0.8702

extraction methods. To evaluate the performance, 80% of reviews are used for training purpose and rest 20% of reviews are used for testing purpose (Tables 3 and 4).

It is obvious from Table 3 that accuracy becomes the lowest 71.35% when BOW has been used for RF classifier along with trigram feature and the highest 87% when ensemble classifier has been applied for the incorporation of unigram and bigram (Unigram + Bigram) features along with BOW technique. For Table 4, f1-score becomes the lowest 71.5% for RF classifier when BOW has been applied along with bigram feature and f1-score reaches a maximum 88.48% when ensemble classifier has been applied for BOW using the incorporation of unigram and bigram (Unigram + Bigram) features.

4.1 Feature

We have considered unigram, bigram, trigram, and incorporation of them, e.g., Unigram + Bigram, Bigram + Trigram, and Unigram + Bigram + Trigram for both TF-IDF and BOW and then applied for RF, SVM, NB, and voting ensemble classifiers in order to investigate classification result.

Unigram (1,1): Firstly, for unigram feature, it has been observed that ensemble classifier achieves the best result rather than all other classifiers by securing accuracy of 86.75% (Fig. 4) and average f1-score of 87.3% for BOW.

Bigram (2,2): For this feature, it has been noticed that ensemble classifier achieves the highest accuracy by securing 80.6% for TI-IDF (Fig. 5) and SVM classifier achieves the highest average f1-score by securing 83.45% for TF-IDF.

Trigram (3,3): This time, trigram feature has been applied and ensemble classifier gains the highest accuracy for TF-IDF by securing 74.75% (Fig. 6) and SVM gains the highest f1-score by securing 82.29% for BOW.

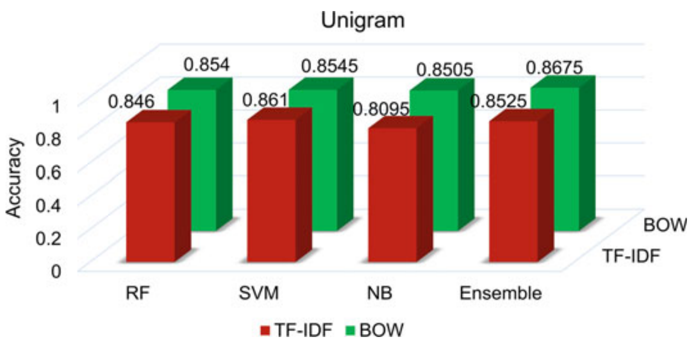

Fig. 4 Accuracy for unigram feature

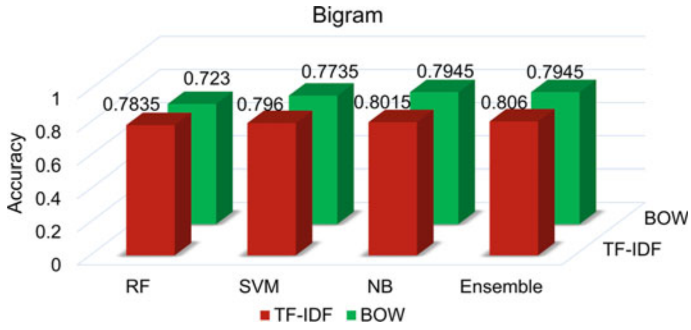

Fig. 5 Accuracy for bigram feature

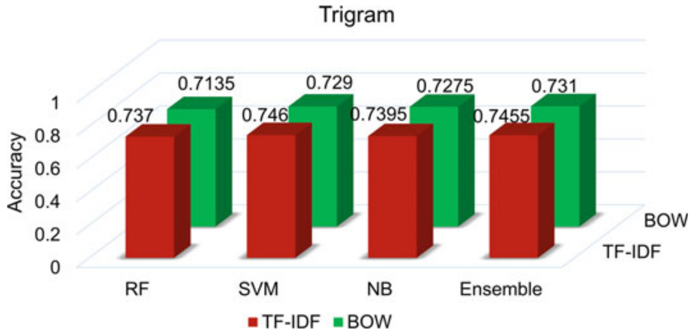

Fig. 6 Accuracy for trigram feature

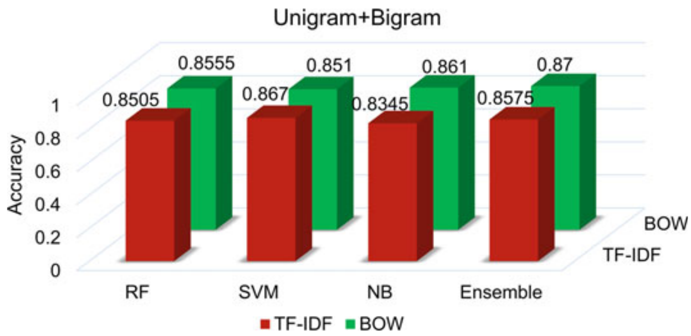

Fig. 7 Accuracy for unigram + bigram feature

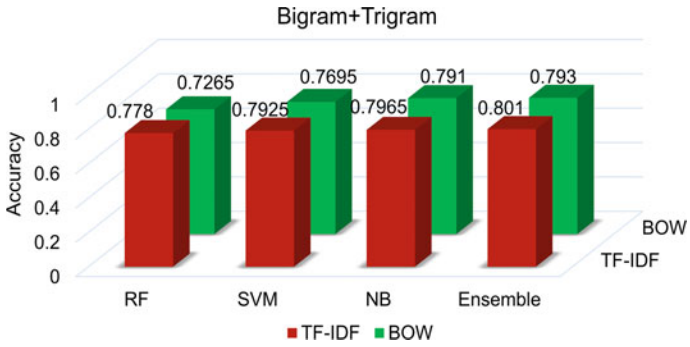

Fig. 8 Accuracy for bigram + trigram feature

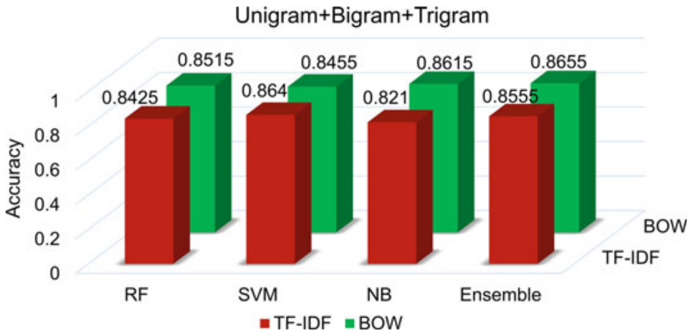

Fig. 9 Accuracy for unigram + bigram + trigram feature

Unigram + Bigram (1,2): When we have used the incorporation of unigram and bigram features, ensemble classifier results in the highest accuracy of 87% for BOW (Fig. 7) and for f1-score, the highest value is 88.48% obtained by ensemble classifier for BOW.

Bigram + Trigram (2,3): For the incorporation of bigram and trigram features, ensemble classifier performs the best accuracy by securing 80.1% for TF-IDF (Fig. 8). For f1-score, ensemble classifier performs the best by securing 83.4% for TF-IDF.

Unigram + Bigram + Trigram (1,2,3): At last, we have applied the incorporation of unigram, bigram, and trigram features and it is notable that ensemble classifier results in the highest accuracy of 86.55% (Fig. 9). For f1-score, the highest score 87.5% which is achieved by ensemble classifier for TF-IDF.

Again, it is very obvious in Fig. 10 that RF classifier performs worse than other classifiers in terms of precision when bigram (2,2), trigram (3,3), and incorporation of bigram and trigram (2,3) features are used separately for TF-IDF technique. From another point of view, in Fig. 11, all the classifiers perform very nearly to each other in terms of precision for BOW technique and any large deviation is not noticeable. From Fig. 12, it is clear that there is no significant deviation in the recall score and

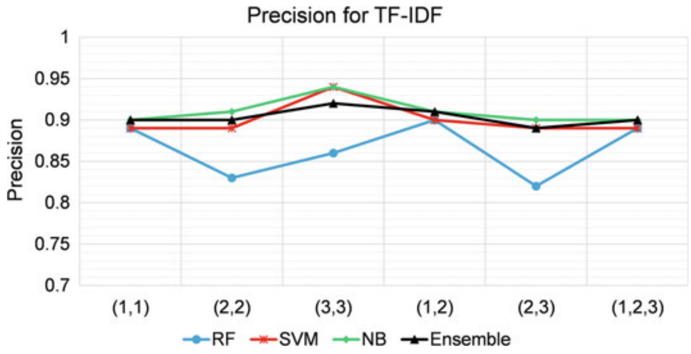

Fig. 10 Average precision for TF-IDF

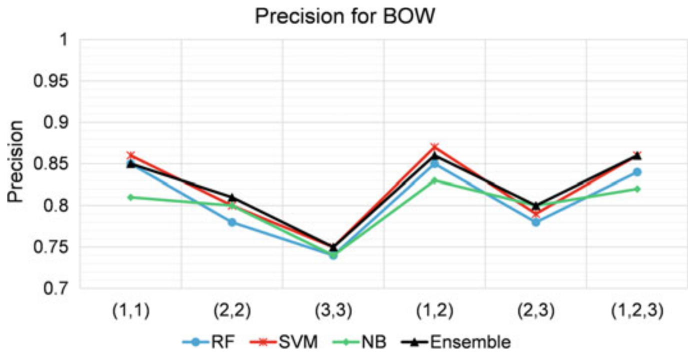

Fig. 11 Average precision for BOW

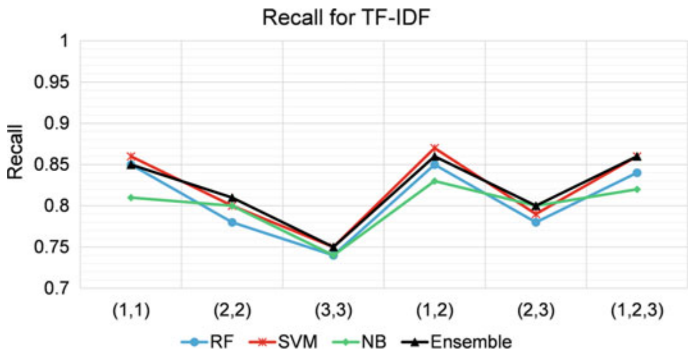

Fig. 12 Average recall for TF-IDF

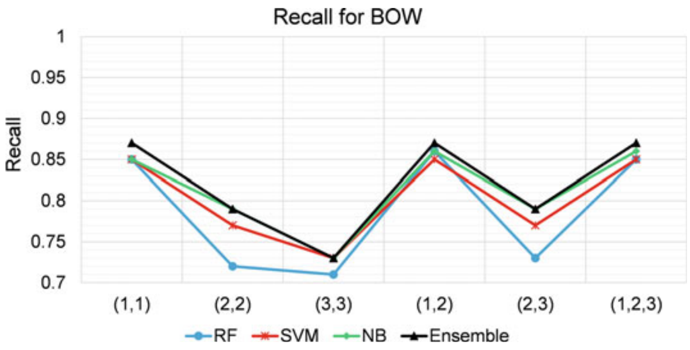

Fig. 13 Average recall for BOW

all classifiers give very close values for TF-IDF. On the other hand, Fig. 13 shows that RF classifier becomes unable to attain better recall score comparing with other classifiers when bigram (2,2), trigram (3,3), and incorporation of bigram and trigram (2,3) features are employed separately for BOW feature extraction technique.

The promising outcomes of our experiment reveal that our proposed approach can perform sentiment classification with good accuracy. There is no doubt that business organizations might be able to identify negative reviews efficiently. After having an inspection over the negative reviews, it is very possible to find out customer demand in a short time and business organizations can reshape their products and policies.

5 Conclusion

In this paper, a powerful system has been introduced for customer reviews sentiment analysis. Our main research objectives have been fulfilled perfectly by the outcomes of our experiment. In our methodology, TF-IDF and BOW feature extraction techniques have been conducted separately for variant N-Gram features and their combination to classify customer reviews. By exploring the outcomes, it is observed that unigram takes advantage of the dataset and boosts up the performance. For higher value of n, the accuracy decreases significantly and the highest accuracy is 87% obtained by voting ensemble classifier for BOW feature extraction method using the amalgamation of unigram and bigram (1,2) features. Again, in the comparison of TF-IDF and BOW techniques, TF-IDF gives better performance for most of the techniques. It is also explored that better accuracy can be achieved through voting ensemble classifier rather than individual classifiers. In future, emoticons that also contain customer sentiment can be considered and hybrid approach may be applied to improve the performance.

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